

Data Warehousing and Data Mining

CSIT — 7th Semester — 2081 — Answer Sheet
Subject Code: CSC410

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define strong association rule. What are the limitations of APriori algorithm? Create a FP tree from the following data set. TIDList of Items T1{A, B, C} T2{B, C, D} T3{C, D} T4{B, D} T5{A, C} T6{A, C, D}

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is the role of Laplace smoothing? Create a decision tree from the following data set using ID3 as attribute selection approach. Object A1 A2 Class 1 TTC12 TTC13 TFC24 FFC15 FTC26 FTC2

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Consider the data set (6,3), (7,2), (4,8), (2,2), (0,2), (9,0). Taking k = 3, show the result after first iteration using k - means algorithm. For choosing initial centroid, use k - means++ by taking (6, 3) as initial cluster center.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Explain about data mining primitives.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define support vector. Write the algorithm for back propagation for classification.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is data mart? Why do we need multidimensional data model?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Describe the different types of data object and attribute types.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is data cube? List the different variations of cube materializations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is the concept behind beam search? Discuss about theory of balance and status.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain about web content, web usage and web structure mining.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Given the following distance matrix, find the core points and outliers using DBSCAN. Take Eps = 2.5 and MinPts = 3.

Data Points	A	B	C	D	E	F	G	H
A	0.1	0.412	0.834	0.245	0.665	0.836	0.405	0.83
B	0.412	0.824	0.244	0.475	0.004	0.47	0.41	2.823
C	0.834	0.244	0.475	0.004	0.47	0.41	2.823	1.63
D	0.245	0.475	0.004	0.47	0.41	2.823	1.63	0.603
E	0.665	0.004	0.47	0.41	2.823	1.63	0.603	0.16
F	0.836	0.405	0.83	0.1	0.412	0.002	0.242	0.00
G	0.405	0.83	2.823	1.63	0.603	0.002	0.242	0.00
H	0.83	2.823	1.63	0.603	0.16	0.00	0.242	0.00

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. List the components of data warehouse. Discuss about the trust propagation on social network.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.